1. Design and explain a three-state divide-by-3 FSM.

2. Write Verilog code for a divide-by-3 FSM and explain the major blocks.

3. Develop the state table for a divide-by-3 FSM.

4. Explain how the divide-by-3 FSM recovers from an invalid state.

5. Compare a divide-by-3 FSM with a simple modulo-3 counter and discuss duty cycle.

6. Design a Mealy FSM to detect the pattern 1011 with overlapping detection. Explain the states and transitions.

7. Write Verilog code for a Mealy FSM that recognizes 1011 with overlapping detection. Explain the major blocks.

8. Develop the state table for a Mealy 1011 pattern recognizer and explain the output.

9. Explain how overlapping detection is implemented in a Mealy FSM for 1011.

10. Compare the hardware and timing characteristics of Mealy and Moore pattern recognizers.

11. Design a Moore FSM to detect the pattern 1011 with overlapping detection. Explain the states and transitions.

12. Write Verilog code for a Moore FSM that recognizes 1011 with overlapping detection. Explain the major blocks.

13. Develop the state table for a Moore 1011 pattern recognizer and explain the output.

14. Explain how overlapping detection is implemented in a Moore FSM for 1011.

15. Compare the hardware and timing characteristics of Moore and Mealy pattern recognizers.

16. Derive the expression for outputs of the half adder and the full adder using SOP from their respective truth tables.

17. Derive the generator and propagator bits as well as the carry out bit for the 4^{th} bit in a 4-bit, in terms of all the lower bits, A_i and B_i where "i" = 3:0 for a carry lookahead adder

18. What is the delay for a the following types of 32-bit adders? Assuming each input gate delay is 120 ps and that a full adder has 300 ps delay

a. **A ripple carry adder**
b. **A carry-lookahead adder with 4-bit blocks**

c. **A prefix adder**

19. Write the VHDL implementation for a 8-bit multiplier using the multiply(*) operation, also write the synthesized circuit for it.

20. Design two adders: a 64-bit ripple-carry adder and a 64-bit carry lookahead adder with 4-bit blocks. Use only two-input gates. Each two-input gate is 15 m2, has a 50 ps delay, and has 20 pF of total gate capacitance. You may assume that the static power is negligible. (a) Compare the area, delay, and power of the adders (operating at 100 MHz). (b) Discuss the trade-offs between power, area, and delay.

21. Explain why a designer might choose to use a ripple-carry adder instead of a carry-look ahead adder.

22. Using the 8-step floating-point addition procedure, compute 6.0 + 2.5, showing the renormalization step, and give the final IEEE 754 encoding.

23. A floating-point addition produces a positive result of magnitude 4.5×10^{38} before storage, exceeding the text's stated maximum normal magnitude of $\pm 3.402824 \times 10^{38}$. Determine (a) whether overflow occurs, (b) what is stored in round-to-nearest mode, and (c) the resulting 32-bit encoding.

24. Using 4 integer bits and 4 fraction bits, compute 2.125 + (−1.8125) in two's complement fixed-point notation, showing all conversion and addition steps.

25. (a) Decode the 32-bit pattern 1 10000101 11010000000000000000000 to its decimal value, showing all steps. (b) Using Table 5.2's special-value rules, state what the patterns 0 00000000 00000000000000000000000 and 0 11111111 00000000000000000000000 each represent.

26. Using 5 integer bits and 4 fraction bits, compute −4.375 + (−2.8125) in two's complement fixed-point notation, showing all conversion and addition steps. Then express the result in sign-and-magnitude notation as well.